



CSE 3433- SOFTWARE ARCHITECTURE

Lecturer: Dr Mohamad Nor Bin Hasan

Topic: Event-Driven Job Recruitment & Application Processing

INDIVIDUAL LOGBOOK

Prepared by : MUHAMAD NUR IMAN HAFIZI BIN MOHD ADIB
(S75659)

System (Event-Driven Architecture)

Group: Group 12

No	Name	Matric Number
1	MUHAMAD NUR IMAN HAFIZI BIN MOHD ADIB	S75659
2	MUHAMMAD HARITH RIEZHAN BIN SYANAZ	S75804
3	MUHAMMAD SYUKRAN BIN MOHD ZAKRI	S75572

Weekly Progress

PHASE 1 (Week 3 - Week 5) - Project Proposal

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 3	Initial project idea discussion and architecture selection	Researched Event-Driven Architecture (EDA) concepts and identified its suitability for recruitment systems. Participated in discussions regarding system scope and recruitment workflow.	Limited understanding of event-driven communication mechanisms and event processing flow.	Study EDA case studies and investigate how modern recruitment platforms process applications through asynchronous events.
Week 4	Requirement gathering and system analysis.	Identified database requirements for candidates, recruiters, job postings, and applications. Contributed to functional requirement discussions	Determining relationships between entities and ensuring data consistency across distributed services.	Review database design principles and analyze recruitment platforms to determine required data structures.
Week 5	Project proposal completion.	Defined database-related functional requirements and supported the preparation of project objectives and system scope.	Ensuring database requirements aligned with the selected Event-Driven Architecture.	Validate requirements against architecture design and review proposal content before submission.

PHASE 2 (Week 6 - Week 7) - Architecture Design

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 6	Study of Event-Driven Architecture and service decomposition.	Designed data storage strategy for Job Management Service and Application Service. Identified database responsibilities for each service.	Understanding database-per-service concepts and event consistency requirements.	Explore microservice database patterns and eventual consistency approaches used in distributed systems.
Week 7	Architecture diagram and service identification development.	Designed database interactions between services and event processing components. Assisted in defining data flow within architecture diagrams.	Mapping database operations to synchronous event flows.	Review event-driven communication examples and refine database interactions in system diagrams.

PHASE 3 (Week 8 - Week 11) - Prototype Progress

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 8	Database design and schema development.	Designed MySQL database schema for recruiters, candidates, job postings, applications, and notifications. Defined primary keys, foreign keys, and table relationships to support recruitment workflows.	Ensuring the database structure could efficiently support multiple recruitment processes while maintaining data consistency.	Review normalization techniques and validate entity relationships before implementation.
Week 9	Database implementation and testing.	Created MySQL tables and tested CRUD operations for job postings, candidate profiles, and application records. Verified referential integrity between related tables.	Managing relationships between multiple tables and avoiding data redundancy.	Optimize table structures and implement constraints to maintain data accuracy.
Week 10	Database integration with system services.	Integrated MySQL database with backend services and tested data retrieval for job search, application submission, and recruitment workflow processes.	Ensuring smooth communication between application services and database operations.	Perform integration testing and optimize SQL queries to improve response time.

Week 11	Database validation and performance review.	Validated database records, tested application tracking functionality, and reviewed query performance for recruitment-related operations. Implemented validation mechanisms to prevent duplicate job applications.	Maintaining data consistency while handling concurrent user requests	Refine database constraints and conduct additional testing to ensure reliability and performance.
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PHASE 4 (Week 12 - Week 14) - Project Report and individual Logbook

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 12	Final report preparation and architecture evaluation.	Prepared database-related sections of the report including Technology Stack, Service Identification, and Team Contributions.	Documenting technical database concepts clearly and consistently.	Follow report template guidelines and perform detailed technical reviews.
Week 13	Final report compilation and logbook preparation.	Reviewed report formatting, validated database descriptions, and completed individual logbook documentation.	Ensuring consistency across diagrams, tables, and technical explanations.	Conduct final proofreading and verify all report sections before submission.

PHASE 5 (Week 14) - Final Demonstration

Week	Activities Undertaken	Individual Contribution	Challenges	Action Plan
Week 14	Project presentation and final demonstration.	Prepared explanations regarding MySQL Database, ElasticSearch functionality, and database architecture decisions. Assisted during system demonstration and technical questioning	Answering technical questions regarding database design and event-driven data consistency	Perform presentation rehearsals and prepare detailed explanations for architecture and database-related decisions.

GenAI Usage Record

WEEK	DATE	TOOL USED	PURPOSE	PROMPT USED	OUTPUT SUMMARY
Week 3	29/03/2026	ChatGPT	Project understanding	"Suggest a software architecture style suitable for a job recruitment system."	Recommended Event-Driven Architecture and explained its benefits.
Week 5	12/04/2026	ChatGPT	Proposal refinement	"Review my project proposal and suggest improvements."	Improved project scope, objectives, and requirement descriptions.
Week 6	19/04/2026	Claude	Architecture Study	"Explain Event-Driven Architecture with practical examples."	Provided detailed explanation of event producers, consumers and event bus concept
Week 7	26/04/2026	Gemini	Database Design	"Design a MySQL database for a recruitment management system"	Generated tables, relationships, and normalization suggestions.
Week 8	03/05/2026	ChatGPT	Service Identification	"Identify services required in an event-driven recruitment system."	Suggested Job Management, Application, Notification, and Search Services.

Week 9	10/05/2026	Microsoft Copilot	Architecture Diagram Support	"Suggest components for an event-driven recruitment architecture diagram.""	Recommended architecture components and service interactions.
Week 10	17/05/2026	Gemini	MySQL Optimization	"How can I optimize MySQL queries for job search operations?"	Suggested indexing strategies and query optimization methods.
Week 11	24/05/2026	ChatGPT	Database design	"Evaluate the strengths and weaknesses of Event-Driven Architecture."	Generated evaluation points related to scalability, performance, and maintainability.
Week 12	31/05/2026	Claude	Technical Writing	"Improve the explanation of deployment architecture for a report."	Enhanced report clarity and technical descriptions.

Week 13	7/06/2026	ChatGPT	Report Compilation	"Help organize software architecture report sections."	Improved report structure and formatting consistency.
Week 14	14/06/2026	Gemini	Presentation Preparation	"What are the key points to present for an Event-Driven Architecture project?"	Generated presentation flow and technical discussion points.